Enning Yang

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ML Platform / MLOps Engineer · Authorized to work in Canada without sponsorship (PR in process)

Summary

Software engineer with 3+ years building and operating machine-learning systems in production. At BrainScores (McGill) I own the stack end to end: a Snakemake/FastAPI MRI-processing platform on an HPC cluster (1,100+ scan sessions, 1,400 tests), its AWS infrastructure (Terraform, ECS Fargate, Cognito, GitHub OIDC CI/CD) and PyTorch training of 3D segmentation and surface models. Maintainer of BrainSpace and BrainStat, open-source Python libraries cited by 800+ papers (~6,800 downloads a month). ICLR 2026 co-author; Nature Communications first author. M.Sc. in Neuroscience & AI, McGill / Mila.

Experience

Research Software Developer (ML Platform & Infrastructure) Sep 2024 – Present
BrainScores (brain-MRI AI startup) · MICA Lab, McGill University, Montreal, QC

Production ML platform and data pipelines

- Built and operate **IronFist**, the lab's MRI-processing platform: a Snakemake workflow running six neuroimaging pipelines (FastSurfer, HippUnfold, micapipe, nnU-Net QC, z-brains) in SHA-pinned Singularity containers on a 256-core SGE cluster, plus a FastAPI run registry, ops dashboard and sign-in portal; 1,100+ sessions, 4 cohorts, 1,394 tests, GitHub Actions CI.
- Doubled cohort throughput (13 → 27 sessions/hour) by re-tuning SGE slot reservation: a 692-session campaign finished in 34 h instead of a projected 3 days with zero failures; fixed 5 faults that only appear at cohort scale.
- Made it safe to run unattended: atomic symlink releases with health-check rollback, auto-resubmission of transient failures, watchdogs, and a SHA-256 lock file (9 components, 2 model weights) checked on every run.
- Built **AutoDICOM**, an hourly scanner-to-BIDS ingestion service (DICOM → NIfTI, atomic publish, never overwrites), in production since Aug 2026: 718 sessions / 518 subjects converted, 0 needing manual attention.
- Rebuilt the lab's 1.4 TB BIDS dataset (43,129 files) non-destructively, taking BIDS-validator errors from 24,915 to 0 with every action logged.
- Agentic workflow: Claude Code and Codex are my primary coding tools, every change gated by CI, tests and code review; 320+ merged pull requests across 27 repositories since 2024.

Cloud, product and web

- Shipped brainscores.ai on AWS (ca-central-1 for Canadian data residency): Next.js 16 app with Cognito sign-in and direct-to-S3 presigned uploads on ECS Fargate behind CloudFront/ALB; all infrastructure in Terraform; GitHub Actions CI/CD with OIDC (no stored keys), main → dev → prod promotion; live since Jul 2026.
- Turned the MICAFlow research pipeline into a self-serve web service: pick a scan, the job runs on the cluster (~40 min) and returns a PDF report plus an interactive 3D viewer (Cornerstone3D, three.js) via an HMAC-signed handoff; Playwright end-to-end, render-safety and left/right tests gate every deploy.

Machine learning

- Trained 3D medical-imaging models with PyTorch DDP on 4× A6000 GPUs: a 66M-parameter residual 3D U-Net with SynthSeg-style domain randomization (13 tissue and pathology classes) and a Neural-ODE mesh-deformation model for cortical surfaces; found an adjoint-gradient bug that had silently frozen the encoder.
- Benchmarked ONNX Runtime / OpenVINO inference for the segmentation and surface models: 2.4× faster on CPU, 2.1× on GPU, bit-identical outputs.

Open-source neuroimaging software (github.com/MICA-MNI)

- Maintainer and release manager of **BrainSpace** (254 stars) and **BrainStat** (119 stars), Python/MATLAB libraries cited by 800+ papers, ~6,800 downloads/month: 14 releases since late 2024 after 2–3 years without one, every open issue closed, CI rebuilt (Python 3.10–3.12, tag-validated PyPI and MATLAB publishing).

- Led the v1 rewrite of **micapipe** (121 stars, cited by 115+ papers) as a Snakemake/snakebids workflow (~500 commits): 8 modules, two-stage Docker/Apptainer builds (the ~50 GB tool image rebuilds in seconds), lint and smoke-test CI on a self-hosted runner.
- Ported C++ tools to Python with parity tests: **zb_command** (27 Connectome Workbench commands, 22 bit-exact, 151 tests, nightly parity CI) and **pyonlm** (MRI denoiser; correlation 1.0000, 1.4× faster).

Founding Software Engineer Mar 2023 – Aug 2024

Eureka Inc. (consumer AI startup), Seattle, WA · remote

- First engineer: designed, built and deployed the backends of two consumer AI products on AWS (ECS Fargate, RDS, ECR, ALB, Route 53) with Terraform and GitLab CI/CD (lint, bandit, pytest, build, deploy).
- Dish-recommendation service (FastAPI, PostgreSQL, SQLAlchemy/Alembic, 47 endpoints): constraint-based recommender with LLM-written explanations (LangChain, Azure OpenAI), exposed as ChatGPT custom-GPT actions; mined ~1,000 Yelp/Google/DoorDash reviews with GPT; A/B-tested.
- Virtual interior-design backend (FastAPI, MongoDB, JWT auth, S3, 42 endpoints): generative-media pipeline with Stable Diffusion text-to-image, Mask2Former + depth-estimation layout generation on SageMaker and Hugging Face endpoints (with failover), MusicGen and GPT object recommendation.

Data Science Intern Jun 2020 – Nov 2020

Intel Corporation, China

- Built Python/Pandas stages of a semiconductor manufacturing-data pipeline used by hardware engineers.

Education

M.Sc. in Neuroscience and Artificial Intelligence Jan 2021 – Nov 2023

McGill University & Mila, Montreal · GPA 3.93/4.0 · UNIQUE ($10K) and Graduate Excellence ($9K) scholarships

B.Eng. in Computer Science (Honors) Sep 2016 – Jul 2020

University of Electronic Science and Technology of China, Chengdu · GPA 4.0/4.0 · Exchange semester: UC Berkeley

Publications & Awards

- **ICLR 2026**, co-author — "Generative Adversarial Post-Training Mitigates Reward Hacking in Live Human-AI Music Interaction." Built the real-time ONNX inference path that lets the model jam live with a human.
- **Nature Communications 2023**, first author — "The default network dominates neural responses to evolving movie stories." NLP and statistical modeling over 50,000+ brain images.
- **Build OS26 AI-agent hackathon** (Mila Ventures, 2026), prize-winning team: AI PI, a Slack multi-agent system in which PI-persona agents review a research proposal like a thesis committee; I built the agent workflows.

Skills

- **Languages:** Python, TypeScript, SQL, Bash, MATLAB, Cython · English, French (B2), Mandarin (native)
- **Machine learning:** PyTorch (DDP), MONAI, nnU-Net, ONNX Runtime, OpenVINO, scikit-learn, Hugging Face; LLM apps and agents: OpenAI/Anthropic APIs, LangChain, AutoGen, Claude Code, Codex
- **Backend and web:** FastAPI, Next.js/React, REST/OpenAPI, PostgreSQL, SQLite, MongoDB, OAuth2/JWT
- **Cloud and DevOps:** AWS (ECS Fargate, ECR, ALB, CloudFront, S3, RDS, Cognito, IAM, SageMaker), Terraform, Docker, Apptainer, GitHub Actions, GitLab CI, OIDC, Linux
- **Pipelines, HPC and quality:** Snakemake, SGE, BIDS, DICOM/NIfTI, FreeSurfer, ANTs, FSL; pytest, Playwright